

Virtual Reality Application That Leverages the Proteus Effect to Make Blood Donation a Positive Experience

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ABSTRACT

In Japan, the reluctance of young people to donate blood has become a social issue. In this study, for the purpose of deterring the trend among young people not to donate blood, we propose a virtual reality (VR) application that changes blood donation into a positive experience by gradually changing the appearance of an avatar from a normal state to a state full of energy according to the elapsed time of blood donation. As a result of a comparison experiment between the proposed method to change blood donation into a positive experience and a method to distract attention by stimulation, it was clarified that the pain of blood donation was significantly alleviated through the proposed method.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Virtual reality.

KEYWORDS

Pain relief, Virtual reality, Proteus effect, Cognitive reappraisal

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1 INTRODUCTION

In Japan, it is becoming a social issue for young people to leave blood donation. According to the Japanese Red Cross Society, the number of blood donors in their 30s or younger decreased by 33% in 2022 compared to 10 years ago, suggesting that if this trend continues, there may be a shortage of blood for transfusion [The Japan Times, 2023]. Young people avoid donating blood because of negative experiences such as the pain of needles [Monobe et al., 2006]. Existing research has developed a VR application [Tarrant et al., 2019] and a mixed reality (MR) application [Pandey et al., 2024] to alleviate the negative experience of blood donation by diverting the blood donor's attention. Alleviating the negative experience of donating blood by turning it into a positive experience is thought

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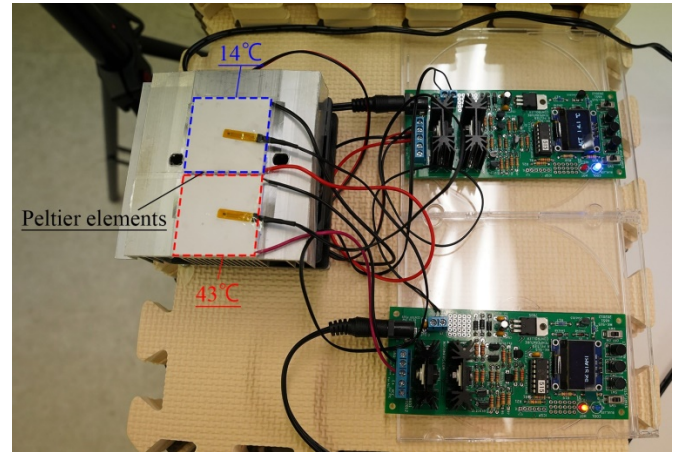


Figure 1: Pain reproduction device.

to be more effective than distraction technique [Webb et al., 2012], but this approach has yet to be developed.

2 PROPOSED METHOD

In this study, we propose a VR application to make blood donation a positive experience by utilizing a psychological phenomenon called the Proteus effect to prevent young people from leaving blood donation. The Proteus effect is a phenomenon in which the recognition of a user is changed by the appearance of an avatar worn by the user in a virtual space. Based on this phenomenon, we seek to alleviate the pain of blood donation by making the user feel that his or her body has been revitalized by donating blood. The system gradually changes the arms and hands of the blood donor's avatar displayed on the head mounted display (HMD) from a normal state to a state of full energy, thereby making the user feel that his or her body has been revitalized by the blood donation. In addition, the sound effects played from the HMD headphones and the heat radiated from the incandescent bulb to the hand are gradually changed in sync with the energy state.

In order to verify the effectiveness of the proposed method, we fabricated a device that reproduces the pain of blood donation using the thermal grill illusion. The thermal grill illusion is a phenomenon in which the brain perceives the illusion of pain by simultaneously presenting hot and cold stimuli to the skin, which alone are not painful. Based on this phenomenon, the device safely reproduces the pain of blood donation by utilizing the exothermic and endothermic reactions of the Peltier element to present hot and cold stimuli (43°C and 14°C) at harmless temperatures to the user's arm (Figure 1).

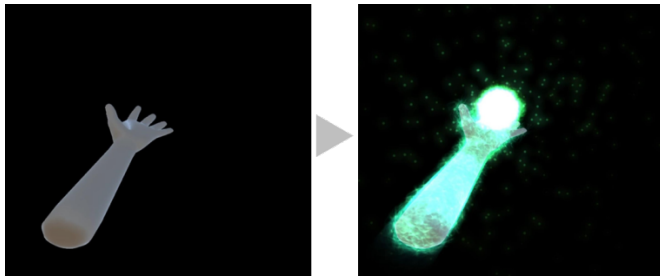


Figure 2: Proposed method.

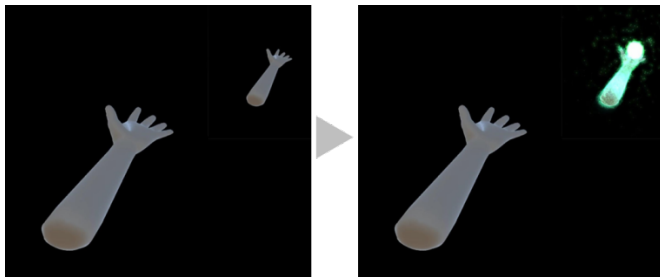


Figure 3: Comparative approach.

3 EXPERIMENT

In this study, 10 healthy adults were subjected to a comparison experiment between the proposed method, which changed blood donation into a positive experience (Figure 2), and a method, which presented another avatar at a distance from the user as a stimulus showing the same visual effect and changed its appearance in the same way as the proposed method to distract attention (Figure 3). The participants experienced VR of a randomly selected method of the above two methods for 5 minutes while their arms were placed on the pain reproduction device (Figure 4). After a 10-minute break, they tried the other method of VR. Questionnaires after the experience of each method included a 5-item version of the Subjective Vitality Scale (SVS), which assesses subjective vitality on a 7-point scale [Kawabata et al., 2016], and the Numerical Rating Scale (NRS), which assesses pain intensity on a 11-point scale. These were used to assess how well the body felt energized and how much pain was relieved.

When the results were analyzed using the Wilcoxon signed-rank test (significance level 5%), the proposed method yielded significantly higher values for SVS ($V = 50.5$, $p = 0.0078$) (Figure 5a). The proposed method also yielded a significantly lower values for NRS ($V = 2.5$, $p = 0.0391$) (Figure 5b). Based on these results, it is considered that the pain of blood donation can be alleviated by feeling that one's body is energized by blood donation through the proposed method.

4 CONCLUSION

In this study, we proposed a VR application that makes blood donation a positive experience by utilizing the Proteus effect. As a result of the comparison experiment with a method to distract attention by stimulation, it was clarified that the pain of blood donation

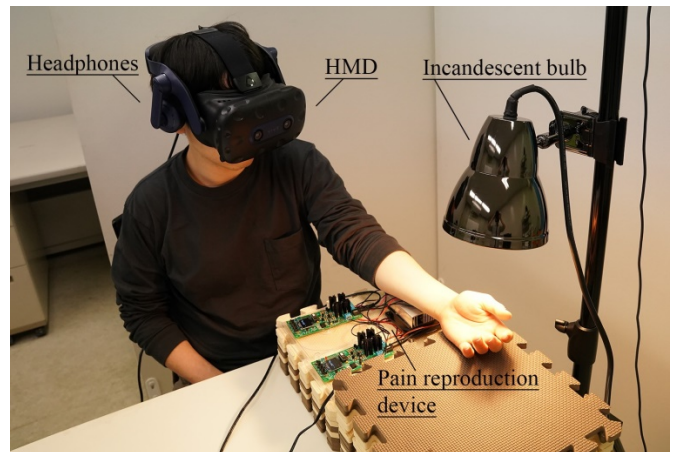


Figure 4: Scene of experiment.

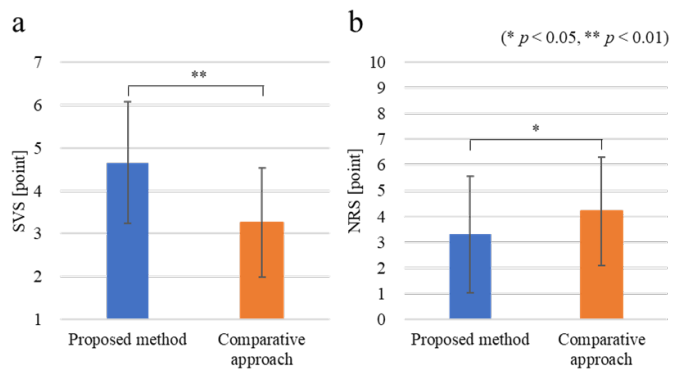


Figure 5: (a) Assessment results of subjective vitality. (b) Assessment results of pain intensity.

was significantly alleviated by the proposed method. As a future prospect, we plan to conduct a demonstration experiment in an actual blood donation situation.

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